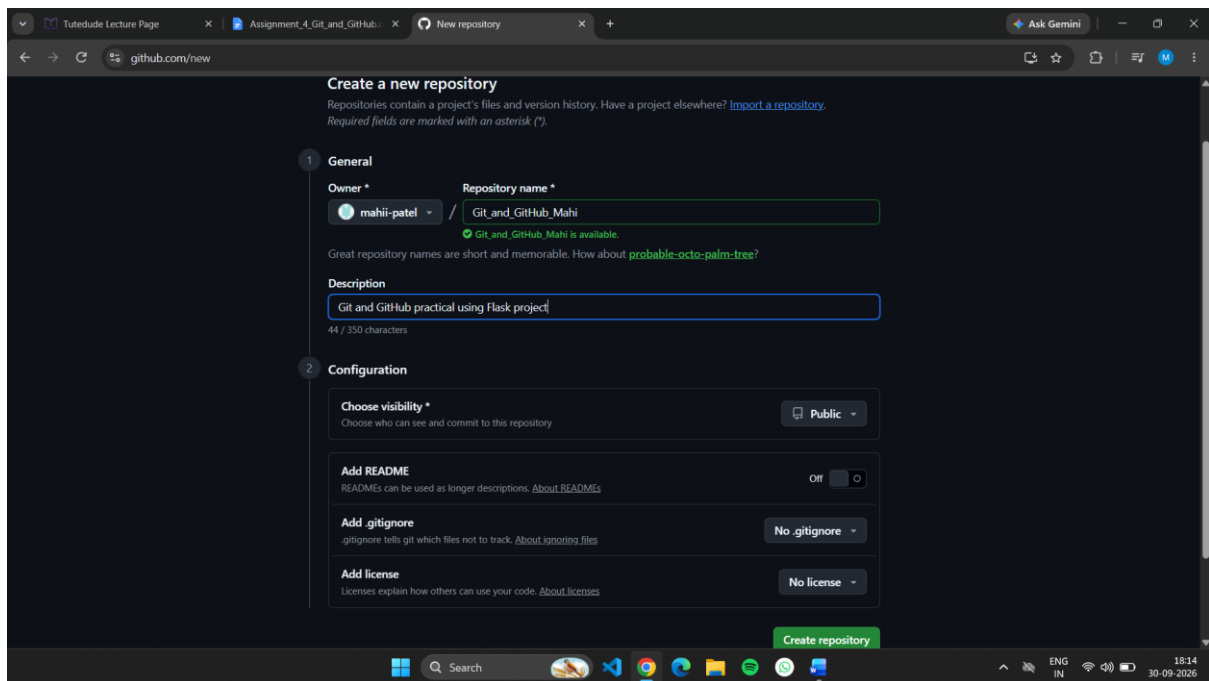


GIT & GITHUB ASSIGNMENT

Task 1: Repository Setup & First Branch

1. Create a new GitHub repository.



2. Clone the repository to your local machine using SSH (generate an SSH key if needed, and add the public key to your GitHub account).

```
MAHI@Mahi MINGW64 ~ (master)
$ ls -al ~/.ssh
total 78
drwxr-xr-x 1 MAHI 197121 0 Sep 28 18:24 ./
drwxr-xr-x 1 MAHI 197121 0 Sep 29 22:37 ../
-rw-r--r-- 1 MAHI 197121 399 May 5 2025 id_ed25519
-rw-r--r-- 1 MAHI 197121 91 May 5 2025 id_ed25519.pub
-rw-r--r-- 1 MAHI 197121 25846 Sep 28 18:24 known_hosts
-rw-r--r-- 1 MAHI 197121 25203 Sep 28 18:24 known_hosts.old

MAHI@Mahi MINGW64 ~ (master)
$ ^C

MAHI@Mahi MINGW64 ~ (master)
$ eval "$(ssh-agent -s)"
Agent pid 1542

MAHI@Mahi MINGW64 ~ (master)
$ ssh-add ~/.ssh/id_ed25519
Identity added: /c/Users/MAHI/.ssh/id_ed25519 (MAHI@Mahi)

MAHI@Mahi MINGW64 ~ (master)
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPOV8PsTFkw3ekp0Zmy70k/Z1hbX50oITsh1gtNmckRs
MAHI@Mahi

MAHI@Mahi MINGW64 ~ (master)
$ ssh -T git@github.com
Hi mahii-patel! You've successfully authenticated, but GitHub does not provide shell access.
```

Git clone:

```

MAHI@Mahi MINGW64 ~/stuff/tutorials (master)
$ git clone git@github.com:mahii-patel/Git_and_GitHub_Mahi.git
Cloning into 'Git_and_GitHub_Mahi'...
warning: You appear to have cloned an empty repository.

MAHI@Mahi MINGW64 ~/stuff/tutorials (master)
$ ^C

MAHI@Mahi MINGW64 ~/stuff/tutorials (master)
$ git clone git@github.com:mahii-patel/Git_and_GitHub_Mahi.git
fatal: destination path 'Git_and_GitHub_Mahi' already exists and is not an empty directory.

MAHI@Mahi MINGW64 ~/stuff/tutorials (master)
$ cd Git_and_GitHub_Mahi

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git remote -v
origin  git@github.com:mahii-patel/Git_and_GitHub_Mahi.git (fetch)
origin  git@github.com:mahii-patel/Git_and_GitHub_Mahi.git (push)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$

```

3. Create a new branch named after your username (e.g., Tutedude).

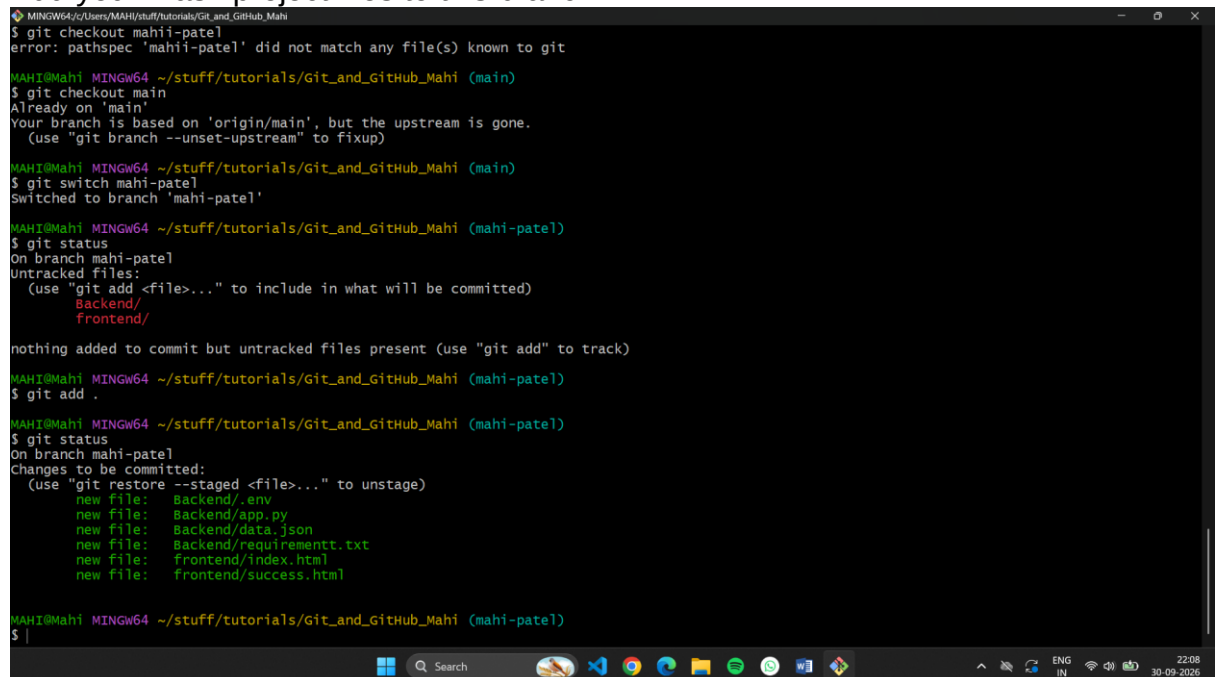
```

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git branch mahi-patel

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git branch
mahi-patel
* main

```

4. Add your Flask project files to this branch.



```

MINGW64/C:/Users/MAHI/stuff/tutorials/Git_and_GitHub_Mahi
$ git checkout mahi-patel
error: pathspec 'mahii-patel' did not match any file(s) known to git

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git checkout main
Already on 'main'
Your branch is based on 'origin/main', but the upstream is gone.
(use 'git branch --unset-upstream' to fixup)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git switch mahi-patel
Switched to branch 'mahii-patel'

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahii-patel)
$ git status
On branch mahii-patel
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Backend/
    frontend/

nothing added to commit but untracked files present (use "git add" to track)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahii-patel)
$ git add .

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahii-patel)
$ git status
On branch mahii-patel
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   Backend/.env
    new file:   Backend/app.py
    new file:   Backend/data.json
    new file:   Backend/requirements.txt
    new file:   frontend/index.html
    new file:   frontend/success.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahii-patel)
$

```

5. Commit the changes and merge the branch into the main branch.

```
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel)
$ git commit -m "Add flask projects files"
[mahi-patel e599deb] Add flask projects files
6 files changed, 125 insertions(+)
create mode 100644 Backend/.env
create mode 100644 Backend/app.py
create mode 100644 Backend/data.json
create mode 100644 Backend/requirements.txt
create mode 100644 frontend/index.html
create mode 100644 frontend/success.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel)
$ git switch main
Switched to branch 'main'
Your branch is based on 'origin/main', but the upstream is gone.
(use 'git branch --unset-upstream' to fixup)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git merge mahi-patel
Updating 20f075a..e599deb
Fast-forward
 Backend/.env          | 1 +
 Backend/app.py        | 49 +++++
 Backend/data.json     | 23 +
 Backend/requirements.txt | 4 +
 frontend/index.html   | 37 +
 frontend/success.html | 11 +
 6 files changed, 125 insertions(+)
create mode 100644 Backend/.env
create mode 100644 Backend/app.py
create mode 100644 Backend/data.json
create mode 100644 Backend/requirements.txt
create mode 100644 frontend/index.html
create mode 100644 frontend/success.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git status
On branch main
Your branch is based on 'origin/main', but the upstream is gone.
(use 'git branch --unset-upstream' to fixup)

nothing to commit, working tree clean

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git branch
  mahi-patel
* main

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$
```

Task 2: Update JSON & Resolve Conflicts

1. Create a new branch named `_new` (e.g., `Tutedude_new`).

```
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git branch mahi-patel-new

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git switch mahi-patel-new
Switched to branch 'mahi-patel-new'

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ git branch
  mahi-patel
* mahi-patel-new
  main

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$
```

2. Update the content of the JSON file used for the `/api` route in this branch.

```
MINGW64/c/Users/MAHI/stuff/tutorials/Git_and_GitHub_Mahi
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ find Backend -type f -name "*.json"
Backend/data.json

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ AC

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git switch mahi-patel-new
Switched to branch 'mahipatel-new'

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahipatel-new)
$ cat AC

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahipatel-new)
$ cat Backend/data.json
[
  {
    "id" : 1,
    "name" : "mahipatel",
    "department" : "Cloud and application development"
  },
  {
    "id" : 2,
    "name" : "Nishra",
    "department" : "Cyber security"
  },
  {
    "id" : 3,
    "name" : "Tirth",
    "department" : "Data Science"
  },
  {
    "id" : 4,
    "name" : "Nishant",
    "department" : "Data Science"
  }
]
```

I update this in vscode

```
File Edit Selection View Go Run Terminal Help
datajson U X
C:\Users\MAHI\Desktop>Flask_and_mongodb_mahi>ASSIGNMENT>Backend>datajson {} 3
1
2
3 {
4   "id" : 1,
5   "name" : "mahipatel",
6   "department" : "Cloud and application development"
7 }
8
9 {
10  "id" : 2,
11  "name" : "Nishra",
12  "department" : "Cyber security"
13 }
14
15 {
16  "id" : 3,
17  "name" : "Tirth",
18  "department" : "Data Science"
19 }
20
21 {
22  "id" : 4,
23  "name" : "Nishant",
24  "department" : "Data Science"
25 }
26
27 {
28   "id" : 5,
29   "name" : "Ishan"
30 }
31 }
```

```
MINGW64/c/Users/MAHI/stuff/tutorials/Git_and_GitHub_Mahi
On branch mahi-patel-new
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Backend/data.json

no changes added to commit (use "git add" and/or "git commit -a")

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ AC

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ git add Backend/data.json

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ git commit -m "update json data"
[mahi-patel-new e3cf125] update json data
1 file changed, 5 insertions(+)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ git push
fatal: The current branch mahi-patel-new has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin mahi-patel-new

To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ git push --set-upstream origin mahi-patel-new
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 12 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (17/17), 2.14 KiB | 1.07 MiB/s, done.
Total 17 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:mahi-patel/Git_and_GitHub_Mahi.git
 * [new branch]      mahi-patel-new -> mahi-patel-new

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ cat Backend/data.json
[
  {
    "id" : 1,
    "name" : "mahi",
    "department" : "Cloud and application development"
  },
  {
    "id" : 2,
    "name" : "Nishra",
    "department" : "Cyber security"
  },
  {
    "id" : 3,
    "name" : "Tirth",
    "department" : "Data Science"
  },
  {
    "id" : 4,
    "name" : "Nishant",
    "department" : "Data Science"
  },
  {
    "id" : 5,
    "name" : "ishan",
    "department" : "Cloud computing"
  }
]
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$
```

3. Merge the _new branch into the main branch.

```

]
}
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ git checkout main
Switched to branch 'main'
Your branch is based on 'origin/main', but the upstream is gone.
(use "git branch --unset-upstream" to fixup)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git merge mahi-patel-new
Updating e599deb..e3cf125
Fast-forward
 Backend/data.json | 5 +++++
 1 file changed, 5 insertions(+)

```

4. If there are conflicts during the merge, resolve them by accepting the changes from the `_new` branch.

```

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:   https://github.com/mahii-patel/Git_and_GitHub_Mahi/pull/new/main
remote:
To github.com:mahii-patel/Git_and_GitHub_Mahi.git
 * [new branch]      main -> main

```

5. Add the resolved changes to the staging area, commit them, and push the updates to the remote repository.

```
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi
{"id": 5,
 "name": "ishan",
 "department": "Cloud computing"
}
]
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (mahi-patel-new)
$ git checkout main
Switched to branch 'main'
Your branch is based on 'origin/main', but the upstream is gone.
(use 'git branch --unset-upstream' to fixup)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git merge mahi-patel-new
Updating e599deb..e3cf125
Fast-forward
 Backend/data.json | 5 +++++
 1 file changed, 5 insertions(+)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:   https://github.com/mahii-patel/Git_and_GitHub_Mahi/pull/new/main
remote:
To github.com:mahii-patel/Git_and_GitHub_Mahi.git
 * [new branch]    main -> main

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ git push origin main
Everything up-to-date

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_GitHub_Mahi (main)
$ |
```

Task 3: Parallel Feature Development

1. **Branch Creation:** Create two branches, master_1 and master_2, from the main branch.

```
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git branch master_1

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git branch master_2

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git branch status

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ ^C

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git branch
  mahi-patel
  mahi-patel-new
* main
  master_1
  master_2
  status

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ |
```

2. **Feature Development in master_1:** Create a To-Do Page in the frontend with a form containing the fields Item Name and Item Description.

```
MINGW64~/Users/MAHI/stuff/tutorials/Git_and_Github_Mahi
$ AC

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ ls frontend
index.html success.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git checkout master_1
Switched to branch 'master_1'

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ ls frontend
index.html success.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ touch frontend/todo.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ ls frontend
index.html success.html todo.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ code frontend/todo.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git status
On branch master_1
```

The image shows a VS Code editor window with a file named `todo.html` open. The file contains the following HTML code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>To-Do Page</title>
7 </head>
8 <body>
9
10  <h1>To-Do List</h1>
11
12  <form>
13    <label for="itemName">Item Name:</label>
14    <input type="text" id="itemName" name="itemName" required>
15
16    <br><br>
17
18    <label for="itemDescription">Item Description:</label>
19    <textarea id="itemDescription" name="itemDescription" required></textarea>
20
21    <br><br>
22
23    <button type="submit">Add Item</button>
24  </form>
25
26 </body>
27 </html>
```

Below the editor, a terminal window shows the following commands and output:

```
MINGW64/c/Users/MAHI/stuff/tutorials/Git_and_Github_Mahi
$ code frontend/todo.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git status
On branch master_1
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        frontend/todo.html

nothing added to commit but untracked files present (use "git add" to track)
MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git add .
MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git commit -m "add to-do page"
[master_1 1618856] add to-do page
1 file changed, 27 insertions(+)
create mode 100644 frontend/todo.html
MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git push --set-upstream origin master_1
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 655 bytes | 655.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'master_1' on GitHub by visiting:
remote:   https://github.com/mahii-patel/Git_and_Github_Mahi/pull/new/master_1
remote:
To github.com:mahii-patel/Git_and_Github_Mahi.git
 * [new branch]      master_1 -> master_1
branch 'master_1' set up to track 'origin/master_1'.
MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$
```

Backend API in master_2: Create a backend route named `/submittodoitem` that accepts `itemName` and `itemDescription` via a POST

```
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi/backend
$ git checkout master_1
Switched to branch 'master_1'
Your branch is up to date with 'origin/master_1'.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git checkout master_2
Switched to branch 'master_2'

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_2)
$ git branch
  master_1
* master_2
status

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_2)
$ code backend/app.py

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_2)
$ cd backend

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi/backend (master_2)
$ python app.py
Traceback (most recent call last):
  File "C:\Users\MAHI\stuff\tutorials\Git_and_Github_Mahi\backend\app.py", line 6, in <module>
    @app.route('/submittodoitem', methods=['POST'])
    ^^^^^
NameError: name 'app' is not defined

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi/backend (master_2)
$ ^C

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi/backend (master_2)
```

```
File Edit Selection View Go Run ...
C: > Users > MAHI > stuff > tutorials > Git_and_Github_Mahi > backend > app.py
7 def home():
8     return render_template("index.html")
9
10
11 @app.route("/todo")
12 def todo():
13     return render_template("todo.html")
14
15
16 @app.route("/submittodoitem", methods=["POST"])
17 def submit_todo_item():
18     item_name = request.form.get("itemName")
19     item_description = request.form.get("itemDescription")
20
21     return {
22         "message": "To-Do item submitted successfully",
23         "itemName": item_name,
24         "itemDescription": item_description
25     }
26
27
28 if __name__ == "__main__":
29     app.run(port=9000, debug=True)
```

Change the form action

```
File Edit Selection View Go Run ... Search
C:\Users\MAHI>stuff>tutorials>Git_and_Github_Mahi>frontend>todo.html>html>body>form
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>To-Do Page</title>
7 </head>
8 <body>
9
10   <h1>To-Do List</h1>
11
12   <form action="/submittodoitem" method="POST">
13     <label for="itemName">Item Name:</label>
14     <input type="text" id="itemName" name="itemName" required>
15
16     <br><br>
17
18     <label for="itemDescription">Item Description:</label>
19     <textarea id="itemDescription" name="itemDescription" required></textarea>
20
21     <br><br>
22
23     <button type="submit">Add Item</button>
24   </form>
25
26 </body>
27 </html>
```

Ln 12, Col 50 (46 selected) Spaces: 4 UTF-8 CRLF HTML Go Live

18:40 01-10-2026

```
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi/backend
mahi-patel
mahi-patel-new
main
master_1
* master_2
status

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_2)
$ code backend/app.py

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_2)
$ cd backend

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi/backend (master_2)
$ python app.py
Traceback (most recent call last):
  File "C:\Users\MAHI\stuff\tutorials\Git_and_Github_Mahi\backend\app.py", line 6, in <module>
    @app.route('/submittodoitem', methods=['POST'])
    ^^^^^
NameError: name 'app' is not defined

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi/backend (master_2)
$ ^C

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi/backend (master_2)
$ python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:9000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 499-008-123
```

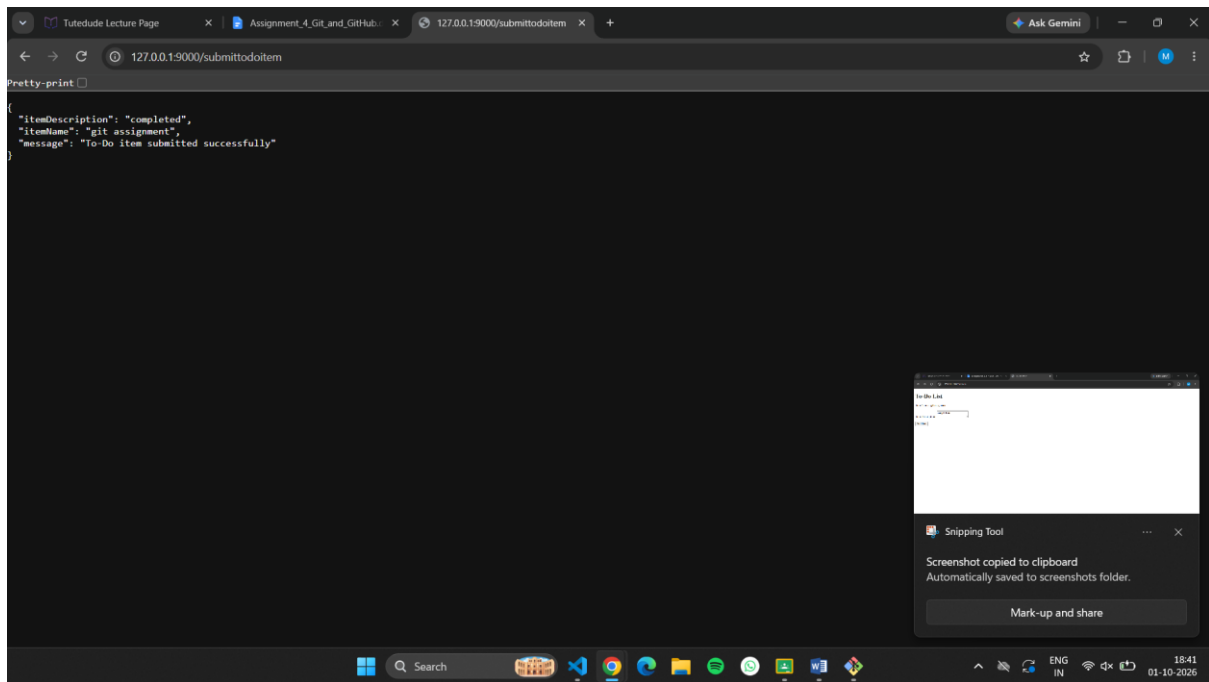
To-Do List

Item Name:

Item Description:

Add Item





4.Merging Changes: Merge the changes from both master_1 and master_2 into the main branch.

```
MINGW64~/c/Users/MAHI/stuff/tutorials/Git_and_Github_Mahi
nothing to commit, working tree clean

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git switch main
Already on 'main'
Your branch is up to date with 'origin/main'.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git pull origin main
From github.com:mahii-patel/Git_and_Github_Mahi
* branch      main      -> FETCH_HEAD
Already up to date.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git merge master_1
Already up to date.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git add frontend/todo.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git commit -m "merge master_1 into main"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git merge master_2
Already up to date.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ ^C

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ |
```

Task 4: Sequential Commits, Reset & Rebase

1. **Enhancing the To-Do Form in master_1:** Add the following fields to the To-Do form: Item ID, Item UUID, Item Hash.

```
MINGW64~/Users/MAHI/stuff/tutorials/Git_and_Github_Mahi
Your branch is up to date with 'origin/master_1'.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git branch
  mahi-patel
  mahi-patel-new
  main
* master_1
  master_2
  status

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ code frontend/todo.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git add frontend/todo.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git commit -m "todo with id uuid"
[master_1 be4248a] todo with id uuid
1 file changed, 23 insertions(+), 20 deletions(-)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git push origin master_1
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 539 bytes | 539.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:mahii-patel/Git_and_Github_Mahi.git
   1618856..be4248a  master_1 -> master_1

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ |
```

```
File Edit Selection View Go Run ... Search
todo.html x app.py
C:\Users\MAHI>stuff>tutorials>Git_and_Github_Mahi>frontend>todo.html
1 <form action="/submittodoitem" method="POST">
2
3   <label for="itemId">Item ID:</label>
4   <input type="text" id="itemId" name="itemId" required>
5
6   <br><br>
7
8   <label for="itemUuid">Item UUID:</label>
9   <input type="text" id="itemUuid" name="itemUuid" required>
10
11   <br><br>
12
13   <label for="itemHash">Item Hash:</label>
14   <input type="text" id="itemHash" name="itemHash" required>
15
16   <br><br>
17
18   <label for="itemName">Item Name:</label>
19   <input type="text" id="itemName" name="itemName" required>
20
21   <br><br>
22
23   <label for="itemDescription">Item Description:</label>
24   <textarea id="itemDescription" name="itemDescription" required></textarea>
25
26   <br><br>
27   <button type="submit">Add Item</button>
28
29
```

2. **Committing in Sequence:** Add and commit each field separately — first commit: Item ID; second commit: Item UUID; third commit: Item Hash.
Item id:

```
File Edit Selection View Go Run ... Search
C:\Users\MAHI> stuff > tutorials > Git_and_Github_Mahi > frontend > <> todo.html > form
1 <form action="/submittodoitem" method="POST">
2
3 <label for="itemId">Item ID:</label>
4 <input type="text" id="itemId" name="itemId" required>
5
6 <br><br>
7
8 <label for="itemName">Item Name:</label>
9 <input type="text" id="itemName" name="itemName" required>
10
11 <br><br>
12
13 <label for="itemDescription">Item Description:</label>
14 <textarea id="itemDescription" name="itemDescription" required></textarea>
15
16 <br><br>
17
18 <button type="submit">Add Item</button>
19
20
21 </form>

Ln 5, Col 1 (99 selected) Spaces: 4 UTF-8 CRLF HTML Go Live
1906 01-10-2026

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git commit -m "todo with id uuid"
[master_1 be4248a] todo with id uuid
1 file changed, 23 insertions(+), 20 deletions(-)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git push origin master_1
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 539 bytes | 539.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:mahii-patel/Git_and_Github_Mahi.git
 1618856..be4248a master_1 -> master_1

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git reset HEAD~1
Unstaged changes after reset:
M frontend/todo.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ code frontend/todo.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git add frontend/todo.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git commit -m "add item id field"
[master_1 7bcd3a0] add item id field
1 file changed, 14 insertions(+), 20 deletions(-)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ |
```

Item uuid:

```
File Edit Selection View Go Run ... Search
C:\Users\MAHI> stuff > tutorials > Git_and_Github_Mahi > frontend > <> todo.html > form
1 <form action="/submittodoitem" method="POST">
2
3
4 <label for="itemUuid">Item UUID:</label>
5 <input type="text" id="itemUuid" name="itemUuid" required>
6
7 <br><br>
8
9 <label for="itemName">Item Name:</label>
10 <input type="text" id="itemName" name="itemName" required>
11
12 <br><br>
13
14 <label for="itemDescription">Item Description:</label>
15 <textarea id="itemDescription" name="itemDescription" required></textarea>
16
17 <br><br>
18
19 <button type="submit">Add Item</button>
20
21 </form>

Ln 6, Col 1 (106 selected) Spaces: 4 UTF-8 CRLF HTML Go Live
1906 01-10-2026

MINGW64~/Users/MAHI/stuff/tutorials/Git_and_Github_Mahi
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 539 bytes | 539.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:mahii-patel/Git_and_Github_Mahi.git
1618856..be4248a master_1 -> master_1

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git reset HEAD~1
Unstaged changes after reset:
M frontend/todo.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ code frontend/todo.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git add frontend/todo.html

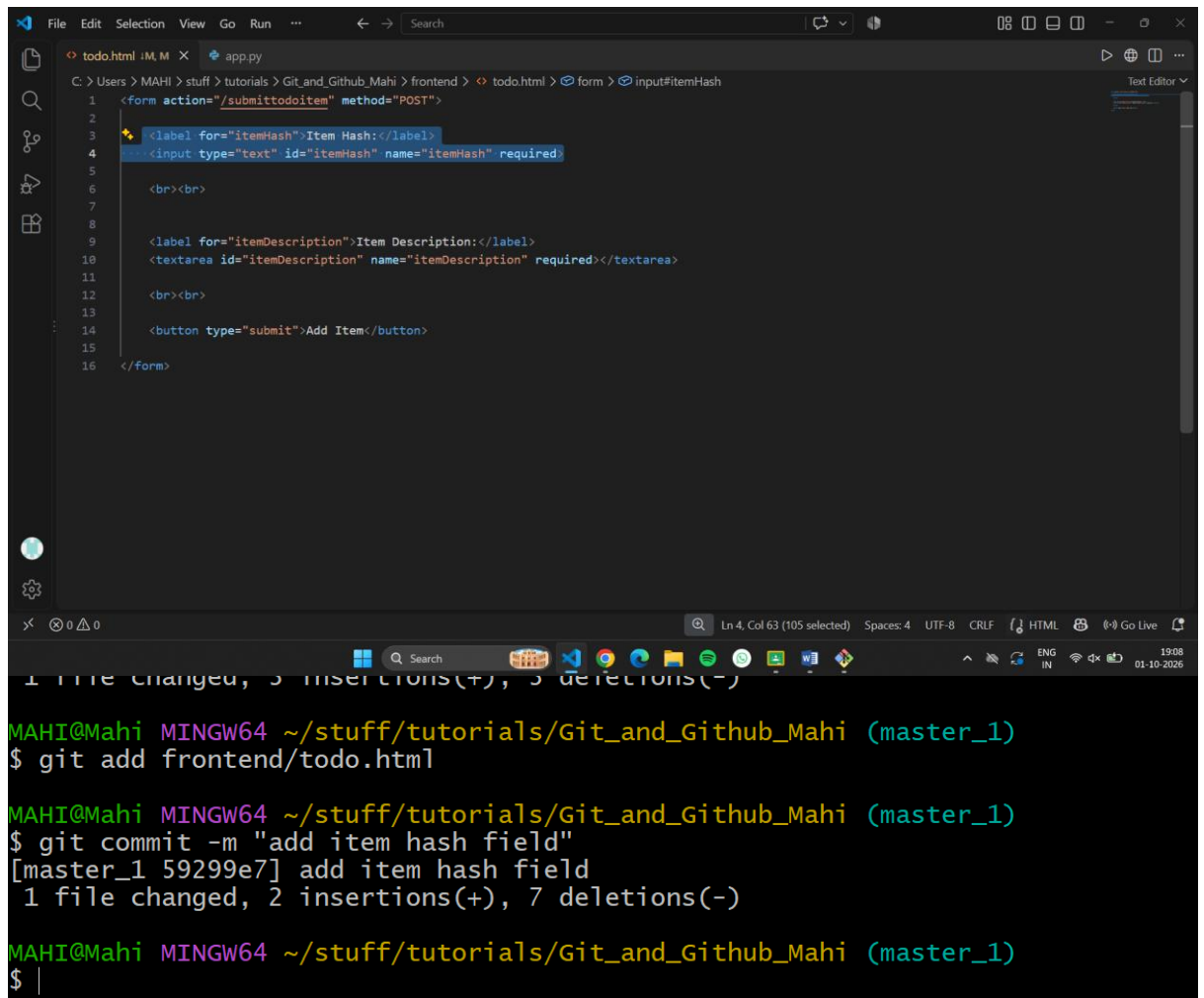
MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git commit -m "add item id field"
[master_1 7bcd3a0] add item id field
1 file changed, 14 insertions(+), 20 deletions(-)

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git add frontend/todo.html

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git commit -m "add uuid i field"
[master_1 d20172b] add uuid i field
1 file changed, 3 insertions(+), 3 deletions(-)

MAHI@MAHI MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$
```

Item hash:



The screenshot shows a Visual Studio Code editor with a file named `todo.html` open. The file contains an HTML form with the following structure:

```
1 <form action="/submittodoitem" method="POST">
2
3   <label for="itemHash">Item Hash:</label>
4   <input type="text" id="itemHash" name="itemHash" required>
5
6   <br><br>
7
8   <label for="itemDescription">Item Description:</label>
9   <textarea id="itemDescription" name="itemDescription" required></textarea>
10
11   <br><br>
12
13   <button type="submit">Add Item</button>
14
15
16 </form>
```

Below the editor, a terminal window shows the following commands and output:

```
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git add frontend/todo.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git commit -m "add item hash field"
[master_1 59299e7] add item hash field
1 file changed, 2 insertions(+), 7 deletions(-)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ |
```

3. **Merging to main:** Merge the master_1 branch into the main branch.

```
MINGW64/c/Users/MAHI/stuff/tutorials/Git_and_Github_Mahi
[master_1 59299e7] add item hash field
1 file changed, 2 insertions(+), 7 deletions(-)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git push origin main
Everything up-to-date

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git merge master_1
Auto-merging frontend/todo.html
CONFLICT (content): Merge conflict in frontend/todo.html
Automatic merge failed; fix conflicts and then commit the result.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main|MERGING)
$ AC

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main|MERGING)
$ git add frontend/todo.html

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main|MERGING)
$ git commit -m "Merge master_1 into main"
[main 56a3a63] Merge master_1 into main

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git push origin main
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 12 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (16/16), 1.44 KiB | 1.44 MiB/s, done.
Total 16 (delta 12), reused 0 (delta 0), pack-reused 0 (from 0)
```

4. Git Reset: In the main branch, roll back to the commit where only the Item ID field was added. Use `git reset --soft` to ensure changes remain staged, then re-commit this state and merge it to main.

```
MINGW64/c/Users/MAHI/stuff/tutorials/Git_and_Github_Mahi
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git log --oneline --decorate
56a3a63 (HEAD -> main, origin/main) Merge master_1 into main
59299e7 (master_1) add item hash field
d20172b add uuid i field
7bcd3a0 add item id field
518a66c Merge master_1 into main
629806a add frontend todo
ae1e7f6 add todo backend
1618856 add to-do page
e3cf125 (origin/mahi-patel-new, status, master_2, mahi-patel-new) update json data
e599deb (mahi-patel) Add flask projects files
a07075a Initial commit

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git reset --soft

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git diff --cached

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git commit -m "Recommit Item ID state"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$
```

5. Rebasing: Rebase the updated main branch changes onto master_1 using `git rebase main master_1`. Preserve individual commits (do NOT squash) to maintain commit history.

```
MINGW64/c:/Users/MAHI/stuff/tutorials/Git_and_Github_Mahi
MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git diff --cached

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git commit -m "Recommit Item ID state"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (main)
$ git rebase main master_1
Successfully rebased and updated refs/heads/master_1.

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git log --oneline --decorate
56a3a63 (HEAD -> master_1, origin/main, main) Merge master_1 into main
59299e7 add item hash field
d20172b add uuid i field
7bcd3a0 add item id field
518a66c Merge master_1 into main
629806a add frontend todo
ae1e7f6 add todo backend
1618856 add to-do page
e3cf125 (origin/mahi-patel-new, status, master_2, mahi-patel-new) update json data
e599deb (mahi-patel) Add flask projects files
a07075a Initial commit

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$ git push --force-with-lease origin master_1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mahii-patel/Git_and_Github_Mahi.git
+ be4248a...56a3a63 master_1 -> master_1 (forced update)

MAHI@Mahi MINGW64 ~/stuff/tutorials/Git_and_Github_Mahi (master_1)
$
```